

Michaela-Ann Mueller

B.S. in Environmental Science

Senior Research Paper

Spring 2026, 0.5 credits

Assessing the Ecological Impacts of Rain Gardens on Soil Health and Vital Ecosystem Functions.

Introduction

Despite a recent global focus on climate change mitigation, green infrastructure and nature-based solutions that are designed to address environmental issues in urban settings remain underresearched and underdeveloped (Sharma & Malaviya, 2021; Su et al., 2024). Among these approaches, rain gardens, also known as bioretention systems, are engineered landscapes that are specifically designed to capture, store, and infiltrate stormwater into the soil while also promoting and restoring ecosystem functions such as carbon sequestration, nutrient cycling, and supporting biodiversity (Bruner et al., 2023; Sharma & Malaviya, 2021).

These systems are emerging technologies in urban sustainability, climate change resilience, and mitigation planning. Rain gardens are shallow vegetated basins that capture stormwater runoff on small scales (Sharma & Malaviya, 2021). However, bioretention systems refer to a much broader category of engineered stormwater infrastructure that may include other constructed features such as underdrains, engineered soils, and internal water storage zones (Öhrn Sagrelius et al., 2022). These strategies combine both intentional vegetation selection and carefully designed landscapes to improve water quality and reduce runoff. Recent literature highlights the rapid rise and expansion of rain garden research and the growing importance of

ecosystem service outcomes, including biodiversity, soil health, and carbon storage (Buzzard et al., 2021; Su et al., 2024). Design analyses further emphasize that soil composition, plant community structure, and hydrologic features, like water storage zones, determine how a system performs and its long-term sustainability (Kravchenko et al., 2024; Öhrn Sagrelus et al., 2022).

In contrast, lawns represent highly degraded and biologically simplified ecosystems. Urban soils are often compacted and lacking organic matter due to intensive land management practices like mowing and fertilization, which reduce infiltration, biological activity, and diversity compared to other natural or restored environments (Pouyat et al., 2002; Wang et al., 2023). Therefore, the extent of damage to the soil structure in lawns makes them an important baseline for comparison because they exemplify the problem that green infrastructure is trying to solve. Moreover, native prairies are an example of the ecological ideal for restored landscapes. Prairies are established plant communities that have deep, complex root structures that help maintain soil aggregation, organic matter storage, and microbial activity (Dutter et al., 2025). By including prairies as a third point of comparison, we can understand how effectively rain gardens restore soil to natural conditions.

Soil health is an essential part of ecosystem functioning and is often evaluated by using indicators like organic matter content, bulk density, infiltration capacity, nutrient content, and microbial activity (Guo et al., 2019). Soil texture, the proportions of sand, silt, and clay, also influence water infiltration, nutrient retention, and the ability for roots to penetrate soil, making it an essential physical characteristic for assessing the quality of soil (Pouyat et al., 2002). These measurements demonstrate the ability of soils to support plant growth, regulate water flow, and cycle nutrients effectively, making them critical for evaluating green infrastructure performance. Soils are also a big part of the global carbon cycle and play an important role in climate

regulation through carbon storage (Lal, 2004). The accumulation of organic carbon in soil happens slowly over time through plant inputs as well as microbial activity and decomposition. Understanding the biological, chemical, and physical properties that make up soil is necessary for assessing ecological impacts and the value of rain gardens for both soil restoration and climate mitigation.

Current field-based hydrological studies show that rain gardens are capable of supporting higher infiltration rates compared to compacted urban soils, exemplifying how vegetation and soil quality interact to regulate water flow (Bortolini & Zanin, 2018; Hess et al., 2021). Improving infiltration also helps reduce pollutant runoff and improve downstream water quality (Wang et al., 2024). Not only do rain gardens provide hydrological benefits, but they also influence carbon sequestration, organic matter accumulation, and nutrient retention. Assessing nutrient dynamics is another important approach for understanding soil functions, since the infiltration of stormwater into soil can alter nitrogen and phosphorus availability and promote organic carbon accumulation, influencing enzyme activity and the efficiency of nutrient cycling (Guo et al., 2019). Studies using lysimeters for soil analysis show that both the layout and soil type used in rain gardens greatly influence nutrient uptake and removal, with newer rain garden systems storing water to enhance nitrogen retention (Tarannum et al., 2025; Wadzuk et al., 2021).

Plant selection is another important aspect of rain garden design that plays a significant role in determining the effectiveness of a rain garden. Studies have shown that both species type and complementarity influence hydrological performance, evapotranspiration, and nutrient uptake, highlighting the ecological function of plant diversity in these systems (Bruner et al., 2023). Furthermore, research comparing native and non-native species of plants indicates that native plants often maintain more stable root structures under fluctuating water conditions than

non-native plants, since they are more adapted to local conditions, allowing them to sustain consistent root growth and help aggregate soil over time (Laukli et al., 2022). Other studies have found that plant structure and morphology affect runoff reduction and suspended solids removal, suggesting a connection between specific plant type and pollutant filtering and infiltration efficiency (Bortolini & Zanin, 2018; Li et al., 2023). These results reinforce the ecological importance of using native plants in campus and community rain gardens.

Over time, rain gardens have the potential to improve the physical structure and integrity of urban soils. One study monitored soils in a newly installed rain garden for a year, finding measurable changes in both microbial activity and soil chemical makeup (Buzzard et al., 2021). This study suggests that bioretention systems can actually accelerate soil formation processes in previously compacted environments. Modeling studies have demonstrated how design features, such as internal water storage zones, help facilitate soil aggregation and microbial development, leading to an improvement in long-term stability and nutrient cycling (Kravchenko et al., 2024; Tarannum et al., 2025). However, it is important to consider that the rate and extent of these benefits depend on plant root depth, soil composition, and the age of a system (Guo et al., 2019). This emphasizes the importance of studying sites with varying ages.

Research on bioretention systems is showing that they may have the ability to partially reverse the degradation of urban soils by restoring porosity, increasing organic matter, and increasing microbial diversity (Buzzard et al., 2021; Guo et al., 2019). Although rain gardens have become increasingly popular in sustainable urban design, studies have mainly examined rain garden performance in isolation or compared to a single land management practice (Buzzard et al., 2021). Understanding these differences is important for determining how effectively rain gardens can facilitate natural ecosystem processes and improve degraded urban landscapes. This

research will contribute to ongoing discussions about the impact of community-based green infrastructure on ecological resilience while also informing future design practices.

This study tests the following hypotheses: (1) Rain gardens planted with native species will have better soil health indicators compared to lawns; (2) Older rain gardens will show greater improvements in soil health indicators than newer ones; (3) Prairie soils will have the highest health indicators across all sites. Rain garden and prairie soils are expected to contain greater organic matter content, lower bulk densities, higher infiltration rates, and increased nutrient retention compared to grass lawns. Since the three rain gardens in this study, located at Seymour Library, Whitcomb Art Center, and Sustainability Cottage, differ in age, from newly installed to approximately 10 years old, this study also predicts that older rain gardens will show greater improvements, but that all rain gardens will still outperform lawns across soil variables. These results are expected due to the ability of native plant communities to foster more complex root structures, supporting soil aggregation and promoting microbial activity and diversity (Bruner et al., 2023; Buzzard et al., 2021). Additionally, rain gardens are designed to capture and infiltrate stormwater efficiently, reducing soil compaction and encouraging soil development (Guo et al., 2019; Hess et al., 2021). Conversely, lawns tend to be intensively managed, degrading soil structure and limiting organic matter accumulation (Li et al., 2023; Sharma & Malaviya, 2021). By explicitly comparing systems of different ages and establishment histories, this study will evaluate whether rain gardens improve soil health relative to lawns and how they compare to long-established native prairie systems (Su et al., 2024; Wang et al., 2023).

Methodology

Sample Sites

This study was conducted on the Knox College campus across seven sites representing three different landcover types with varying levels of management (Figure 1).

Soil Quality Study Sites at Knox College

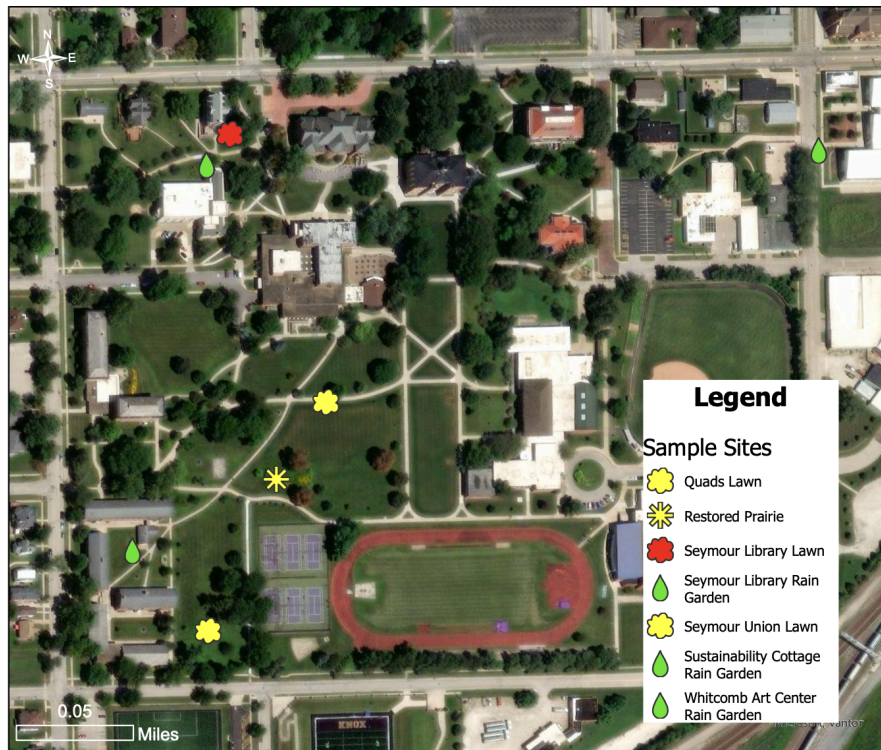


Figure 1. ArcGIS Pro map of soil quality study sites at Knox College. Shows the spatial distribution of soil compaction levels across all seven sites. Tree symbols represent lawns, droplet symbols represent rain gardens, and the asterisk represents the prairie site. Each color is associated with a range of compaction from low to high (green = < 150 psi, yellow = 150 - 250 psi, red = > 350 psi).

Rain Gardens - engineered bioretention systems planted with native species

North side of Seymour Library - Established September 2017

West entrance of Whitcomb Art Center - Established in the summer of 2016

South side of the Sustainability Cottage - Established July 2025

Prairie - a restored native prairie ecosystem

South side of Seymour Union - Established in 2004

Lawns - managed grass areas located near other sites

North side of Seymour Library

East side of the Quads

South side of Seymour Union

Field Sampling

Permission for sampling was obtained from Patrick Pendergast, Head of Campus Grounds. Field sampling was conducted on April 9th, 2026, for six of the seven sites. The WAC Rain Garden samples were taken on April 10th. All samples were taken at a core depth that ranged from 8 to 18 cm, depending on soil compaction and site conditions. At each site, three soil cores were collected at random locations to account for variability within each site. Two cores were collected using a metal coring rod and transferred into soil sample bags, while the third core was collected using a metal corer with a plastic tube inserted to maintain core integrity for bulk density analysis. All cores were approximately 2.1 cm in diameter. Due to soil compaction at several sites, all cores collected were small, rather than the original plan of two small cores and one large core. At each location, GPS coordinates were recorded, and digital photographs were taken to document site characteristics, including vegetation and ground cover.

Sample Preparation and Laboratory Analysis

Two of the three cores collected at each site were combined to create a composite sample. Approximately 10 grams of each composite sample was reserved for pH analysis in the Knox College Environmental Science Laboratory (SMC D219). The remaining sample was then

portioned into seven sample bags filled one-third of the way full. These were sent to AgSource Laboratories for soil analysis, including organic matter, pH, and nutrient levels including base saturation and cation exchange capacity. The third core, collected in the plastic tube insert, was kept intact for bulk density and loss-of-ignition analysis.

pH Analysis - Conducted on April 10th, 2026. Approximately 10 grams of composite soil was combined with 10 mL of deionized (DI) water in a 200 mL beaker and thoroughly mixed. A calibrated pH probe was rinsed with DI water, inserted into the soil mixture, and left to stabilize before reading. The probe was rinsed with DI water between samples.

Bulk Density - Conducted on April 10th, 2026. The intact core in the plastic tube was used for this analysis. The volume of the core cylinder was calculated based on its dimensions. The wet sample was weighed and oven-dried for 24 hours at 105 degrees C, then reweighed. Bulk density was calculated as the dry mass of the soil divided by the volume of the soil core.

Loss-of-Ignition - Conducted on April 17th, 2026. The soil samples used for Bulk Density were crushed slightly and stored in the desiccator between analyses. Approximately 20 grams of oven-dried soil was placed in a ceramic crucible and combusted for 2 hours at 360 degrees C. Samples were cooled in the desiccator and reweighed. The percent organic matter was calculated as the loss of mass during combustion relative to the initial oven-dried mass (Adelsberger, 2025).

Soil Texture Analysis - Conducted on April 28th and 29th, 2026. Approximately 40 grams of oven-dried soil from each of the seven sites was weighed and treated with 10 mL of 30% hydrogen peroxide at a time until frothing subsided. Each beaker was then placed on a hot plate at 90 degrees Celsius for 45 minutes. After cooling for an hour, 100 mL of Calgon solution was mixed into the beaker and allowed to soak overnight. The following day, samples were

transferred to 1000 mL graduated cylinders filled to volume with DI water. One blank solution (Calgon solution + distilled water) was prepared alongside the samples. A hydrometer was used to measure suspension density at 40 seconds and after 6 hours. The temperature was recorded at each reading. Sand percentages were calculated from the 40-second reading, clay from the 6-hour reading, and silt from the difference. The WAC Rain Garden sample experienced mass loss due to overflow during the hydrogen peroxide pre-treatment.

Compaction Testing - Conducted on May 7th, 2026. Soil compaction resistance was measured at two random locations within each of the seven sites using a hand penetrometer. Readings were recorded as resistance ranges (100-200 psi - low compaction, 200-300 psi - moderate compaction, and >300 psi - high compaction). Collected at the same general area as cores.

Spatial Analysis - ArcGIS Pro was used to visualize all seven sites on a map of the campus. Coordinates were compiled into an Excel spreadsheet along with all other data collected for each site and converted to a comma-separated values file. The .csv file was then imported into ArcGIS Pro and georeferenced using the XY coordinate data to create points for each site location on an aerial map of Knox College. The point symbols were then customized to display unique values. Symbol shapes were assigned according to landcover type. The color was assigned according to soil compaction category (<150 psi = green for low compaction, 150-250 = yellow for moderate compaction, >250 = red for high compaction) based on the penetrometer readings.

Data Analysis - All statistical analyses were performed using a one-way analysis of variance (ANOVA) with Tukey-Kramer post-hoc tests to assess differences among landcover types and rain garden ages. This particular approach was chosen because it allows for

comparison across all groups. All analyses were performed using the ANOVA spreadsheet tool from the Handbook of Biological Statistics (McDonald, 2015). Data were analyzed without transformation, and the statistical significance was set at $\alpha = 0.05$ for all tests. Statistical significance was determined according to the overall ANOVA test. Tukey-Kramer post hoc tests were only interpreted for the variables that were statistically significant ($p\text{-value} < 0.05$) due to small sample sizes.

Data Visualization - Microsoft Excel was used to create three tables summarizing the numerical data and statistical results for soil physical and chemical properties, soil nutrients, and nutrient saturation and cation exchange capacity for all seven sites. These tables were then transferred to Google Sheets for formatting. The raw data from laboratory analyses and field measurements were organized by site location and variable type. Bar charts with custom standard deviation error bars were created in Microsoft Excel to visualize key results. For each figure, means were calculated using the AVERAGE function, and standard deviations were calculated using the STDEV.S function.

Analysis I: Landcover Type Comparison

The first analysis tested whether soil health indicators differed significantly among the three landcover types: rain gardens, prairie, and lawns. A one-way ANOVA was conducted for each soil variable measured, with landcover type as the independent variable. Variables tested included bulk density, soil texture (sand, silt, clay as percentages), organic matter content, pH, nutrients (phosphorus, potassium, magnesium, calcium, sodium, sulfur, zinc, manganese, copper, iron, and boron), cation exchange capacity, base saturation, and percent saturation values for cations (hydrogen, potassium, magnesium, calcium, and sodium). Soil compaction was assessed

by converting field measurements (recorded as pressure ranges) to numerical midpoint values (150 - mildly compact, 250 - moderately compact, and 350 - highly compact psi) to allow for statistical analysis.

Sample sizes varied depending on the variable. For most variables, rain gardens had $n = 3$, prairie had $n = 1$, and lawns had $n = 3$. For variables measured at both Knox College and by AgSource (pH and organic matter), rain gardens had $n = 6$, prairie had $n = 2$, and lawns had $n = 6$. When there were significant differences in the ANOVA, Tukey-Kramer tests were used to identify which landcover types differed significantly.

Analysis II: Rain Garden Age Comparison

A second round of analysis was conducted to examine whether soil health indicators differed between established (approximately 10 years old - $n = 2$) and recently installed (approximately 1 year old - $n = 1$) rain gardens. A one-way ANOVA was conducted for each of the same soil variables from the first analysis.

Results

Overview of Soil Variables and Data

Soil physical, chemical, and nutrient properties measured across all seven sites are given in Tables 1, 2, and 3, along with p-values for both landcover and rain garden age analyses. Table 1 focuses on physical and chemical properties, including bulk density, compaction, texture, organic matter, and pH. Table 2 presents the macronutrients (phosphorus, potassium, magnesium, calcium) and the micronutrients (sodium, sulfur, zinc, manganese, copper, iron, boron). Table 3 includes nutrient saturation percentages for hydrogen, potassium, magnesium, calcium, and

sodium, along with base saturation and cation exchange capacity. Of the variables analyzed using one-way ANOVA tests, only a small number showed statistically significant results. For landcover type comparisons, this included compaction, magnesium content, and magnesium and potassium percent saturation. For rain garden age comparisons, only silt and iron content were found to be statistically significantly different.

Sampling Site	Bulk Density (g/cm ³)	% Sand	% Silt	% Clay	Compaction (psi)	Organic Matter (%)	pH
Seymour Library Rain Garden	1.421	54.0905	42.6482	3.2613	150	5.8159	8.25
					150	5.1	7.9
Whitcomb Art Center Rain Garden	1.2357	57.0193	42.7308	0.2499	150	6.9197	7.97
					150	4.2	7.5
Sustainability Cottage Rain Garden	1.2857	42.1677	56.8352	0.9971	150	9.1195	8.21
					350	53.6	7.2
Restored Prairie	1.2088	41.8155	57.6851	0.4994	250	6.523	8.06
					250	3.5	7.1
Seymour Library Lawn	1.2117	24.3909	71.3388	4.2703	350	6.4384	8.43
					350	6.7	6.8
Quads Lawn	1.4763	53.4546	46.0449	0.5005	250	3.4741	8.17
					350	5.3	7.2
Seymour Union Lawn	0.942	77.5331	21.4684	0.9985	250	14.2372	8.01
					350	4.9	5.4
Landcover - p value	0.803	0.903	0.866	0.805	*0.015	0.571	0.596
Age - p value	0.835	0.119	*0.003	0.82	0.178	0.131	0.632

Table 1. Soil Physical & Chemical Properties. Values for bulk density, texture, compaction, organic matter, and pH across rain gardens, prairie, and lawns. Landcover comparison: Rain Garden n = 3, Prairie n = 1, Lawn n = 3. Age comparison: Established Rain Garden n = 2, New Rain Garden n = 1. Group number doubles in compaction, organic matter, and pH due to

replicate measurements. P-values from one-way ANOVA are shown; asterisks indicate the level of statistical significance (*p < 0.05, **p < 0.01).

Sampling Site	P	K	Mg	Ca	Na	Su	Zn	Mn	Cu	Fe	B
Seymour Library Rain Garden	23	115	199	3184	6.6	2.8	7.5	3.3	1.3	28.3	0.4
Whitcomb Art Center Rain Garden	74	219	191	4262	13.3	9.6	33	6.1	2.5	27.3	1.3
Sustainability Cottage Rain Garden	6	73	124	1521	5.1	1.1	1	2.8	0.4	14	0.5
Restored Prairie	41	218	445	2245	10.5	3.1	13.8	1.8	2	28.8	0.9
Seymour Library Lawn	32	244	299	2004	20.2	3.7	9.2	0	0	0	0
Quads Lawn	65	342	362	2694	10.9	4.5	34.1	1.3	4.6	26.3	0.9
Seymour Union Lawn	19	278	342	1566	9.9	6.6	6.8	3.3	1.4	114.6	0.3
Landcover - p value	0.975	0.101	**0.005	0.601	0.498	0.897	0.973	0.282	0.914	0.797	0.602
Age - p value	0.512	0.486	0.062	0.255	0.557	0.546	0.544	0.577	0.386	*0.04	0.731

Table 2. Soil Nutrients. All values given in parts per million (ppm). Values for soil macro and micro nutrients. Landcover comparison: Rain Garden n = 3, Prairie n = 1, Lawn n = 3. Age comparison: Established Rain Garden n = 2, New Rain Garden n = 1. P-values from one-way ANOVA are shown; asterisks indicate the level of statistical significance (*p < 0.05, **p < 0.01).

Sampling Site	% H	% K	% Mg	% Ca	%Na	Base Saturation	Cation Exchange Capacity
Seymour Library Rain Garden	0.1	1.6	9.2	89	0.2	99.9	17.9

Whitcomb Art Center Rain Garden	0	2.4	6.7	90.6	0.2	100	23.5
Sustainability Cottage Rain Garden	0.1	2.1	11.6	85.9	0.3	99.9	8.8
Restored Prairie	0.1	3.6	23.6	72.4	0.3	99.9	15.5
Seymour Library Lawn	13.2	4.1	16.2	65.9	0.6	86.8	15.2
Quads Lawn	0.1	5.1	17.2	77.5	0.3	99.9	17.4
Seymour Union Lawn	43.5	3.5	14	38.8	0.2	56.6	20.1
Landcover - p value	0.383	*0.032	**0.008	0.167	0.605	0.383	0.947
Age - p value	0.667	0.909	0.341	0.217	/0 error	0.667	0.246

Table 3. Nutrients percent saturation, base saturation, and cation exchange capacity.

Percent saturation for hydrogen, potassium, magnesium, calcium, and sodium, along with base saturation and cation exchange capacity values. Landcover comparison: Rain Garden n = 3, Prairie n = 1, Lawn n = 3. Age comparison: Established Rain Garden n = 2, New Rain Garden n = 1. P-values from one-way ANOVA are shown; asterisks indicate the level of statistical significance (*p < 0.05, **p < 0.01).

Landcover Type Comparison

One-way ANOVA tests were performed to compare soil health indicators across three landcover types: rain gardens, lawns, and prairie (Table 1). Soil compaction varied significantly among landcover types (p = 0.015), with rain gardens having the lowest average compaction

(183.33 psi), followed by the prairie site (250 psi), and lawns (316.67 psi) (Figure 1 & 2).

Tukey-Kramer tests indicated that rain gardens had significantly lower compaction than lawns.

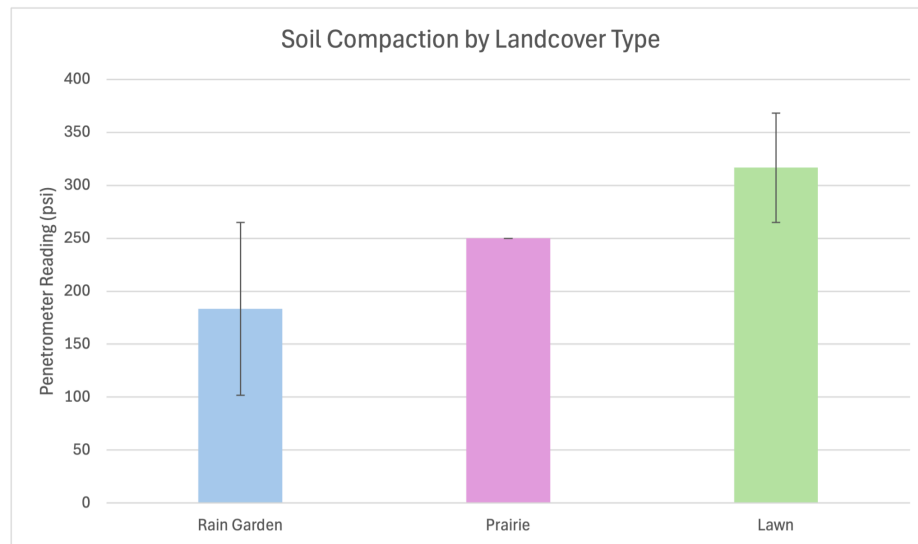


Figure 2. Mean soil compaction in psi across landcover types. Error bars show the standard deviation. Rain gardens had statistically significantly less compaction than lawns ($p = 0.015$).

The distribution of compacted soils in all seven sites is shown in Figure 1. In the map, each symbol shape represents a landcover type and each color indicates the level of compaction. Rain gardens were consistently the least compacted. The prairie was moderately compacted along with two of the lawn sites, and the third lawn was highly compacted. This supports the hypothesis that rain gardens help improve the physical structure of soil compared to lawns (Figure 1).

Table 2 summarizes the soil nutrients readings and statistical results. Many of the nutrient analyses showed no significant differences among landcover types. The macronutrients, such as phosphorus and potassium, did not differ significantly ($p = 0.975$, $p = 0.101$). Organic matter also did not show any significant variability according to landcover type ($p = 0.571$). However, magnesium content differed significantly among landcover types ($p = 0.005$). The prairie site had

the highest magnesium content (445 ppm), followed by the lawn sites (334 ppm), and the rain garden sites had the lowest (171 ppm)(Figure 3). Tukey-Kramer analysis showed that all three landcover types differed significantly from each other.

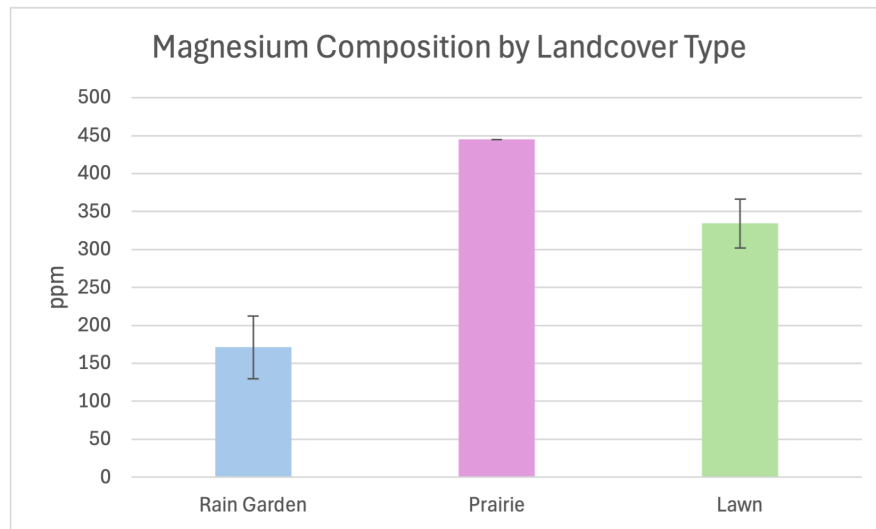


Figure 3. Average magnesium content in parts per million (ppm) for each landcover type. Error bars represent the standard deviation. Magnesium content differed significantly among all three landcover types ($p = 0.005$).

The percent saturation of magnesium also varied significantly among landcover types ($p = 0.008$) (Table 3, Figure 3). The percent saturation of potassium differed significantly, too ($p = 0.032$) (Table 3). Other nutrients, including calcium, sodium, sulfur, zinc, iron, and boron, did not show any statistically significant differences in the ANOVA tests (Table 2). No significant differences were found between landcover types when analyzing the texture components, sand, silt, and clay for the landcover comparison (Table 1). Similarly, bulk density and pH had no significant differences among landcover types. Furthermore, percent saturation for hydrogen, calcium, and sodium, along with base saturation and cation exchange capacity, showed no significant differences across rain garden, prairie, and lawn sites (Table 3).

Rain Garden Age Comparison

In order to investigate differences in rain garden soil health indicators according to age, one-way ANOVAs were used to compare established rain gardens (>10 years, Seymour Library and Whitcomb Art Center) to a newly installed rain garden (<1 year, Sustainability Cottage). Here, soil texture did show a significant difference in silt content ($p = 0.003$). The established rain gardens had less silt (42.69%) compared to the new one (56.84%) (Figure 4). Sand and clay percentages were not significantly different from each other between the two groups.

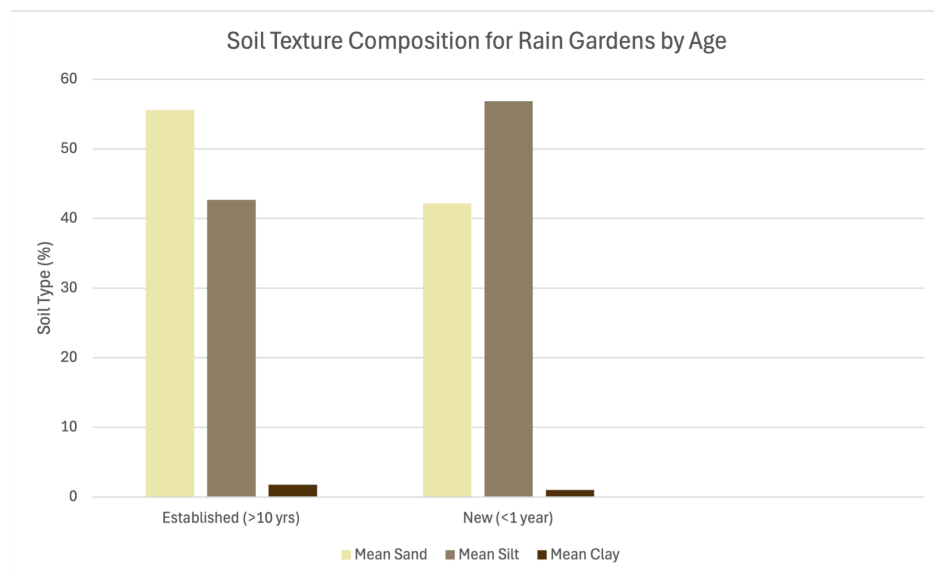


Figure 4. Soil texture composition (% sand, silt, clay) in established (>10 years) and new (<1 year) rain gardens. Silt content varies significantly between the two groups ($p = 0.003$).

Most other soil health indicators did not show statistically significant differences between established and recently installed rain garden sites. Bulk density, organic matter content, pH, and compaction all exhibited no significant differences. Among nutrients analyzed, only iron content was significantly different ($p = 0.040$) (Table 2). Established rain gardens had higher iron content (27.8 ppm) compared to the new rain garden (14 ppm) (Table 2).

Discussion

This study looked at soil health indicators across rain gardens, lawns, and prairie ecosystems on the Knox College campus. The goal was to evaluate whether rain gardens improved soil quality compared to managed lawns and see how they compare to a restored prairie on campus. Some aspects of the results did support the initial hypotheses, including improvements in soil compaction in rain gardens. However, there were not many significant improvements across all variables.

One noteworthy finding was the significant reduction in soil compaction in rain gardens compared to lawns (Figure 1). This supports the hypothesis that rain gardens planted with native species will exhibit better soil health indicators than lawn sites. Some causes for this improvement may include the deeper and more complex root systems present in native plant communities, opposed to turfgrass, which helps penetrate and aerate soil (Bruner et al., 2023). Also, rain gardens are constructed to capture stormwater, creating a cycle of wetting and drying that promotes soil aggregation and reduces density (Guo et al., 2019). Field observations confirm that the rain gardens are successfully capturing stormwater runoff during precipitation events. Another explanation is that rain gardens are not managed through frequent mowing and maintenance, which compacts and degrades soil structure (Pouyat et al., 2002).

The moderate compaction observed in the restored prairie was unexpected, given that prairies are often considered an ecological ideal with deep root structures that maintain soil health (Dutter et al., 2025). However, there are several factors that could account for this outcome. First, the history of the site prior to the restoration in 2004 could impact the soil today. Secondly, prairie ecosystems are more natural, unmanaged ecosystems with higher species diversity and more variability in root structures compared to rain gardens, which are intentionally

designed with hand-picked native species optimized for stormwater infiltration. Additionally, field observations revealed that some areas of the prairie now have non-native species, which likely alter the soil's structure. However, the prairie still showed lower compaction than lawn sites, suggesting that native plant communities show improvements over non-native lawn grasses regardless of whether they are engineered or natural systems.

Surprisingly, there were no significant differences in organic matter content between landcover types, contrary to the hypothesis that rain gardens would have a higher organic matter content than lawns. This was unexpected, especially due to the findings given by Buzzard et al. (2021), where measurable changes in both soil chemistry and biological activity were present after just one year of rain garden establishment. There are many factors that could explain this result. There was a small sample size ($n = 3$ for rain gardens and lawns, $n = 1$ for prairie), which substantially limited the statistical power to detect meaningful differences, especially given the high within-site variability shown by large standard deviations. Furthermore, sampling may have introduced bias since some sites were easier to access than others topographically, and soil cores were more easily obtained from exposed patches than densely vegetated areas. Therefore, more uniform lawns may have provided more representative samples compared to rain garden and prairie sites. Another consideration is that the Sustainability Cottage installation included a mulch layer that may have contaminated soil samples and elevated organic matter measurements, reducing differences in new rain gardens and older sites. Likewise, sampling in early April, before many plants actually begin growing, could have decreased the organic matter present in the soil. However, according to the Knox College Grounds Department, the lawn areas sampled have not been fertilized since at least 2023-2024, which makes recent fertilization an unlikely explanation for the lack of differences in organic matter content.

Similarly, most macronutrients were not significantly different according to landcover type. Since the lawn sites that were sampled have not been fertilized recently, lawn management practices are not likely the cause of elevated nutrient concentrations. Plus, nutrient cycling could be more active in rain garden and prairie sites, resulting in similar readings across landcover types despite different nutrient inputs (Guo et al., 2019; Wadzuk et al., 2021). Magnesium content being highest in the prairie site may reflect differences in plant community composition (Figure 3). The prairie has a more diverse group of plants, likely with a variety of nutrient uptake and root depths compared to the rain gardens, which were intentionally selected for stormwater management. Different plant species accumulate and cycle nutrients at different rates, and prairie plants could potentially access magnesium from deeper soil horizons (Dutter et al., 2025). The increased magnesium content in the prairie soils might be a reflection of the deeper root systems and greater plant diversity characteristics of prairie ecosystems. Prairies can have roots that extend many feet into the ground, allowing them to access and cycle nutrients from deeper soil horizons that are not available to lawn grasses or younger, less established plants in the rain gardens (Dutter et al., 2025). Therefore, it is possible that the diversity of plant species results in different nutrient cycles that could concentrate magnesium in the surface soil.

The comparison between new and established rain gardens showed a significant difference in silt content, with established rain gardens containing less silt than the new ones (Figure 4). This partially supports the hypothesis that older rain gardens would show greater improvements in soil health. However, soil texture differences between established and new rain gardens ($p = 0.003$ for silt) likely reflect variations in parent material at different locations on campus rather than changes over time due to rain garden systems, since soil texture is determined by the parent material and is usually stable over time (Pouyat et al., 2002). However, established

root structures may play some role in transporting fine silt particles deeper into the soil, reducing silt concentrations in the surface horizon sampled. In addition, repeated saturation and draining in rain gardens may promote more aggregation of small particles like silt and clay (Guo et al., 2019).

Overall, most soil health indicators did not differ significantly between established and new rain gardens, suggesting that structural benefits of rain gardens may occur relatively quickly, while other chemical improvements may take much longer to accumulate or need larger sample sizes to detect. There could be other site-specific factors influencing soil health, including drainage area and stormwater input. For example, the Sustainability Cottage rain garden is designed to collect much less water compared to the Whitcomb Art Center and Seymour Library gardens.

When interpreting the results of this study, it is important to keep in mind some of the potential limitations. Firstly, the small sample size considerably limits the reliability of the statistical analyses and the ability to generalize the results. Ideally, multiple replicates would have strengthened the site comparisons. The unequal sample sizes also meant that the ANOVA was more sensitive to variation within some sites than others. Future studies should include more replicates. Secondly, sampling biases may have had an effect on the results. Since soil cores were collected more easily from bare patches, the vegetated areas that likely have more organic matter and biological activity were underrepresented. This bias would disproportionately impact the rain garden and prairie sites compared to the easily accessible lawns. Third, the sampling core depth varied greatly among sites depending on compaction and the presence of rocks or other obstructing objects. While all sites did have variation in compaction and soil conditions, the variability in core sample depths may have introduced more room for uncertainty into the data.

Future studies should use a standardized core depth for sampling to improve consistency across sites.

Another shortcoming of this study was the estimation of penetrometer readings and the conversion from categorical ranges to midpoints for statistical analysis. Also, the Whitcomb Art Center rain garden sample lost mass during the soil texture analysis due to overflow during the hydrogen peroxide pretreatment, although the data were used in the analysis. This methodological issue likely introduced error into the texture measurements for that site. Furthermore, this study did not directly measure infiltration due to site and equipment challenges, which would have provided a better understanding of physical soil functionality. Finally, the Whitcomb Art Center rain garden has a very different design and composition of both soil and plants compared to the other two rain gardens. Additionally, the small sample sizes, particularly the prairie ($n = 1$), severely limited statistical power to detect differences. While there were trends observed in several of the variables (phosphorus, potassium, and other nutrients), these did not reach statistical significance in the overall ANOVA tests and should be interpreted with caution.

Despite these limitations, this study presents valuable evidence that rain gardens can improve soil physical properties in urban settings, especially through reducing soil compaction. The difference in compaction between rain gardens and managed lawns suggests that soil structure benefits can be realized within a decade. This has important implications for stormwater management, since low compaction directly increases the soil's infiltration capacity, reduces runoff, and improves water quality (Hess et al., 2021; Wang et al., 2024).

For those considering installing rain gardens, this study indicates that rain gardens improve stormwater management and soil structure relatively quickly. However, other ecosystem

services develop more gradually or may require larger systems to be detectable. Redirecting stormwater seems to be the main immediate benefit, which aligns with the design intent of bioretention systems (Sharma & Malaviya, 2021). Moreover, rain gardens are not only valuable for their functionality but also for their aesthetic and educational contributions to sustainability. At Knox, there remains interest among both faculty and students in expanding rain garden installations for these very reasons.

Future research should consider several important questions raised by this study. First, monitoring soil development in rain gardens over long periods of time with larger sample sizes would give a clearer picture of soil health improvements over time and help identify which indicators develop the quickest. Next, sampling throughout all seasons could capture changes in soil dynamics that may have been missed by sampling in early spring. Third, more random sampling that ensures a representative portion of each site is covered, potentially using systematic grid sampling, could reduce bias and improve the reliability of the data collected. Fourth, measuring infiltration rates can help link physical characteristics of soil to functional outcomes and verify whether reduced compaction actually improves infiltration. Finally, investigating the history of the sites sampled can help explain the changes or lack thereof that are observed in soil health.

Conclusion

Overall, this study demonstrated that rain gardens reduce soil compaction compared to lawns, partially supporting the hypothesis that rain gardens improve soil health indicators. However, rain gardens did not show significant improvements in organic matter content or most nutrients, contrary to expectations based on previous studies and literature. These results likely

reflect a combination of small sample size, methodological limitations, sampling bias, site-specific variability in design and water inputs, and the possibility that chemical changes develop over longer timeframes.

Age-related differences in soil texture and nutrients like iron suggest that some soil developmental processes continue to modify rain garden soils over time, but long-term research with larger sample sizes is needed to fully understand this. While the prairie was less compacted than lawns, it did not show the lowest compaction or highest organic matter content as hypothesized. This suggests that rain gardens with more intentionally selected plant communities and soil media may reach similar or even better soil outcomes compared to prairies.

These findings support the implementation of rain gardens for urban stormwater management with realistic expectations that functional benefits related to soil physical structure and infiltration capacity are likely obtained within the first few years, while more comprehensive restoration, including accumulating organic matter and cycling nutrients, likely requires longer timeframes or may differ depending on site-specific factors and conditions. It is important to recognize rain gardens' multiple purposes, including stormwater management, aesthetic value, and educational opportunities. This study highlights the need for continued monitoring of rain gardens to fully understand their long-term impacts on soil health.

Bibliography

- Adelsberger, K. (2025). Loss of Ignition Handout. Unpublished laboratory protocol, Knox College, Galesburg, IL.
- Bortolini, L., & Zanin, G. (2018). Hydrological behaviour of rain gardens and plant suitability: A study in the Veneto plain (north-eastern Italy) conditions. *Urban Forestry & Urban Greening*, 34, 121–133. (131251094). <https://doi.org/10.1016/j.ufug.2018.06.007>

- Bruner, S. G., Palmer, M. I., Griffin, K. L., & Naeem, S. (2023). Planting design influences green infrastructure performance: Plant species identity and complementarity in rain gardens. *Ecological Applications*, 33(7), 1–15. (172991227). <https://doi.org/10.1002/eap.2902>
- Buzzard, V., Gil-Loaiza, J., Graf Grachet, N., Talkington, H., Youngerman, C., Tfaily, M. M., & Meredith, L. K. (2021). Green infrastructure influences soil health: Biological divergence one year after installation. *Science of The Total Environment*, 801, 149644. <https://doi.org/10.1016/j.scitotenv.2021.149644>
- Dutter, C. R., Rieke, E. L., Pottebaum, S., & McDaniel, M. D. (2025). Prairie strips improve many measures of soil health in nearly a decade. *Journal of Soil and Water Conservation*, 80(4), 388–404. <https://doi.org/10.1080/00224561.2024.2435683>
- Guo, C., Li, J., Li, H., & Li, Y. (2019). Influences of stormwater concentration infiltration on soil nitrogen, phosphorus, TOC and their relations with enzyme activity in rain garden. *Chemosphere*, 233, 207–215. (137454397). <https://doi.org/10.1016/j.chemosphere.2019.05.236>
- Hess, A., Wadzuk, B., & Welker, A. (2021). Evapotranspiration estimation in rain gardens using soil moisture sensors. *Vadose Zone Journal*, 20(1), e20100. <https://doi.org/10.1002/vzj2.20100>
- Kravchenko, M., Trach, Y., Trach, R., Tkachenko, T., & Mileikovskyi, V. (2024). Improving the Efficiency and Environmental Friendliness of Urban Stormwater Management by Enhancing the Water Filtration Model in Rain Gardens. *Water*, 16(10), 1316. <https://doi.org/10.3390/w16101316>
- Lal, R. (2004). Soil Carbon Sequestration Impacts on Global Climate Change and Food Security. *Science*, 304(5677), 1623–1627. <https://doi.org/10.1126/science.1097396>
- Laukli, K., Gamborg, M., Haraldsen, T. K., & Vike, E. (2022). Soil and plant selection for rain gardens along streets and roads in cold climates: Simulated cyclic flooding and

- real-scale studies of five herbaceous perennial species. *Urban Forestry & Urban Greening*, 68, 127477. <https://doi.org/10.1016/j.ufug.2022.127477>
- Li, C., Chen, L., Li, H., Miao, Z., Yang, R., Chu, L., & Mao, L. (2023). Effects of different types of plants on runoff and reduction and suspended solids removal in rain gardens. *Journal of Environmental Engineering & Landscape Management*, 31(2), 113–120. (164692659). <https://doi.org/10.3846/jeelm.2023.19016>
- McDonald, J. H. (2015, July 20). *One-way anova—Handbook of Biological Statistics*. Sparky House Publishing. Handbook of Biological Statistics. <https://biostathandbook.com/onewayanova.html>
- Öhrn Sagrelus, P., Blecken, G., Hedström, A., Ashley, R., & Viklander, M. (2022). Environmental impacts of stormwater bioretention systems with various design and construction components. *Journal of Cleaner Production*, 359, 132091. <https://doi.org/10.1016/j.jclepro.2022.132091>
- Pouyat, R., Groffman, P., Yesilonis, I., & Hernandez, L. (2002). Soil carbon pools and fluxes in urban ecosystems. *Environmental Pollution*, 116, S107–S118. [https://doi.org/10.1016/S0269-7491\(01\)00263-9](https://doi.org/10.1016/S0269-7491(01)00263-9)
- Sharma, R., & Malaviya, P. (2021). Management of stormwater pollution using green infrastructure: The role of rain gardens. *WIREs Water*, 8(2), e1507. <https://doi.org/10.1002/wat2.1507>
- Su, J., Wang, M., Zhang, D., Sun, C., Zhao, X., & Razi, M. A. B. M. (2024). A systematic and bibliometric review of bioretention system (BRS) for urban ecosystem regulation services. *Urban Climate*, 55, 101923. <https://doi.org/10.1016/j.uclim.2024.101923>
- Tarannum, T., Adhikari, B., & McPhillips, L. E. (2025). Internal Water Storage as a Means of Improving Nitrogen Retention and Reducing Greenhouse Gas Emission in Stormwater Treatment. *ACS ES&T Water*, 5(3), 1137–1147. <https://doi.org/10.1021/acsestwater.4c00864>

Wadzuk, B., DelVecchio, T., Sample-Lord, K., Ahmed, M., & Welker, A. (2021). Nutrient Removal in Rain Garden Lysimeters with Different Soil Types. *Journal of Sustainable Water in the Built Environment*, 7(1), 04020018.

<https://doi.org/10.1061/JSWBAY.0000924>

Wang, M., Sun, C., & Zhang, D. (2023). Opportunities and challenges in green stormwater infrastructure (GSI): A comprehensive and bibliometric review of ecosystem services from 2000 to 2021. *Environmental Research*, 236, 116701.

<https://doi.org/10.1016/j.envres.2023.116701>

Wang, M., Zhuang, J., Sun, C., Wang, L., Zhang, M., Fan, C., & Li, J. (2024). The Application of Rain Gardens in Urban Environments: A Bibliometric Review. *Land*, 13(10), 1702.

<https://doi.org/10.3390/land13101702>